

AD23B31 - Image Processing and Computer Vision

Project Title

A MINI PROJECT REPORT

**Face Mask Compliance Detection
using Computer Vision and Deep Learning**

LBP + HOG + SVM

Submitted by

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MINI PROJECT REPORT

On

Face Mask Compliance Detection

using Computer Vision and Image Processing

Individual Report — Algorithm: LBP + HOG + SVM Classifier

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TABLE OF CONTENTS

1. Introduction	3
1.1 Background and Motivation	3
1.2 Problem Statement	3
1.3 Objectives	3
1.4 Scope of the Report	4
2. Literature Review	4
2.1 Traditional Feature-Based Methods	4
2.2 Deep Learning Approaches	5
2.3 Gap Identified	5
3. Dataset Description	6
3.1 Kaggle Face Mask Dataset	6
3.2 Data Split	6
4. Preprocessing Methodology	7
4.1 Resizing	7
4.2 Gaussian Denoising	7
4.3 CLAHE Contrast Enhancement	7
4.4 Before vs After Visualisation	8
5. Algorithm: LBP + HOG + SVM	8
5.1 Algorithm Overview	8
5.2 Stage 1 — LBP Feature Extraction	8
5.3 Stage 2 — HOG Feature Extraction	9
5.4 Stage 3 — SVM Classification	10
5.5 Implementation Details	10
5.6 Algorithm Flowchart	11
6. Results and Analysis	11
6.1 Quantitative Results	11
6.2 Effect of Preprocessing	12
6.3 Confusion Matrix Analysis	12
7. Discussion	13
8. Conclusion and Future Work	14
References	15
Appendix A — Source Code	16

LIST OF TABLES AND FIGURES

Tables

- Table 1: Dataset Statistics
- Table 2: Train / Test Split
- Table 3: LBP Parameter Configuration
- Table 4: HOG Parameter Configuration
- Table 5: SVM Implementation Configuration
- Table 6: LBP + HOG + SVM Test Set Results
- Table 7: Preprocessing Ablation Study

Figures

- Figure 1: Dataset class distribution (bar chart)
- Figure 2: Before vs After preprocessing — 3-stage comparison
- Figure 3: LBP texture visualisation
- Figure 4: HOG gradient visualisation
- Figure 5: Algorithm architecture flowchart
- Figure 6: Confusion matrix heatmap
- Figure 7: Real-time detection output screenshots

1. INTRODUCTION

1.1 Background and Motivation

The spread of airborne respiratory diseases has made face mask compliance a critical public health requirement across the globe. Ensuring that individuals in public spaces, healthcare facilities, offices, and educational institutions consistently wear masks is a significant challenge for human supervisors — particularly in high-traffic environments where manual monitoring is impractical at scale.

Traditional compliance monitoring relies on human personnel physically observing individuals or reviewing CCTV footage — both approaches are reactive, labour-intensive, and inconsistent. The emergence of affordable high-resolution camera infrastructure combined with advances in computer vision and machine learning has created a practical opportunity for automated, real-time mask compliance detection.

This project implements a complete Face Mask Compliance Detection system using a classical computer vision pipeline: Local Binary Pattern (LBP) and Histogram of Oriented Gradients (HOG) for feature extraction, combined with a Support Vector Machine (SVM) for binary classification. This approach was chosen over deep learning to demonstrate a lightweight, interpretable, CPU-deployable solution requiring no GPU acceleration.

1.2 Problem Statement

Public spaces, healthcare environments, and institutions require continuous face mask compliance monitoring. Manual inspection methods are inefficient and unreliable at scale. Despite established health guidelines, ensuring consistent mask-wearing remains a major operational challenge.

Head and respiratory protection through face masks significantly reduces disease transmission risk. Yet real-world compliance monitoring systems are either too expensive (deep learning requiring GPU hardware) or too slow (manual observation). With advances in computer vision, there is a clear opportunity to automate this detection using lightweight, CPU-efficient classical methods.

Therefore, there is a need for a robust and efficient automated system that can detect whether a person is wearing a face mask, operating in real time using standard computing hardware.

1.3 Objectives

The specific objectives of this project are:

1. To design and implement a complete image preprocessing pipeline (Resize, Gaussian Denoising, CLAHE contrast enhancement) tailored to face detection imagery.
2. To extract discriminative LBP texture features and HOG gradient features from preprocessed face images and analyse their combined feature dimensionality.
3. To train a Support Vector Machine (SVM) classifier with RBF kernel on the combined LBP+HOG feature vectors, with StandardScaler normalisation.
4. To evaluate the trained model on a held-out test set using accuracy, precision, recall, F1-score, and confusion matrix metrics.
5. To develop a complete real-time prediction system supporting live webcam input using OpenCV Haar Cascade face detection integrated with the trained SVM.
6. To document all results, discuss limitations, and propose directions for future improvement.

1.4 Scope of the Report

This report presents the complete end-to-end pipeline, from raw dataset collection and preprocessing through to model development and real-time webcam-based deployment. The system focuses specifically on binary classification: determining whether a person is wearing a face mask (with_mask) or not (without_mask).

The implementation is carried out using Python, leveraging OpenCV for image processing and real-time detection, scikit-image for LBP feature extraction, and scikit-learn for SVM training and evaluation. The system is designed to operate entirely on classical computer vision and machine learning techniques, without deep learning frameworks, ensuring a lightweight CPU-efficient solution.

2. LITERATURE REVIEW

Face mask detection in public health contexts has gained significant attention, particularly since 2020. This section reviews key prior works relating to classical feature-based approaches and deep learning alternatives.

2.1 Traditional Feature-Based Methods

Before the deep learning era, object detection and classification systems relied heavily on hand-crafted feature descriptors. The most influential for human-centric detection is the Histogram of Oriented Gradients (HOG) descriptor, introduced by Dalal and Triggs (2005) for pedestrian detection. HOG captures the distribution of local intensity gradients, providing robust shape representation that is largely invariant to small photometric transformations. Combined with an SVM classifier, HOG established itself as a foundational computer vision method.

The Local Binary Pattern (LBP) descriptor, introduced by Ojala et al. (2002), provides complementary texture information by encoding the relationship between a pixel and its surrounding neighbourhood. LBP has been widely applied to face recognition and texture classification tasks, making it a natural choice for analysing facial regions including mask boundaries.

Combining LBP and HOG has been shown in the literature to outperform either descriptor used alone, as LBP captures fine-grained texture while HOG captures structural edge and shape information — both relevant to distinguishing masked from unmasked faces.

2.2 Deep Learning Approaches

The introduction of deep Convolutional Neural Networks (CNNs) fundamentally changed computer vision. For face mask detection specifically, several deep learning approaches have been proposed. RetinaFace combined with MobileNetV2 has achieved over 94% accuracy on balanced mask datasets. More recent YOLO-based approaches push accuracy above 97% with real-time inference.

However, deep learning carries significant deployment drawbacks in resource-constrained environments: GPU hardware is required for real-time inference, large labelled datasets are needed for training, and the models are difficult to interpret. These limitations motivate continued study of classical methods such as LBP+HOG+SVM for lightweight deployable solutions.

2.3 Gap Identified

A review of existing literature reveals key gaps that this project addresses:

Gap 1 — Complete classical pipeline: Most published mask detection studies use deep learning. This project validates a complete LBP+HOG+SVM classical pipeline achieving 90.21% accuracy on a large real-world dataset of 7,553 images — competitive with classical baselines reported in the literature.

Gap 2 — Combined feature approach: Most classical studies use either LBP or HOG in isolation. This project demonstrates that combining LBP texture features with HOG gradient features improves accuracy from 70.55% (LBP alone) to 90.21% (LBP+HOG combined).

Gap 3 — CPU-deployable real-time system: This project validates that an LBP+HOG+SVM pipeline running entirely on CPU achieves 90.21% accuracy with real-time webcam inference, making it viable for deployment without specialised hardware.

3. DATASET DESCRIPTION

3.1 Kaggle Face Mask Dataset

The Face Mask Dataset used in this project was sourced from Kaggle (Omkar Gurav). It contains real-world photographs of people with and without face masks under various lighting conditions, backgrounds, angles, and demographics. The dataset provides a practical foundation for training and evaluating the mask classification system.

Table 1: Dataset Statistics

Category	Number of Images	Description
with_mask	3,725	Persons wearing face masks
without_mask	3,828	Persons without face masks
Total	7,553	Combined dataset

3.2 Data Split

The dataset was split into training and test sets using an 80/20 stratified split to maintain class distribution balance across both subsets. Stratification ensures that both classes are proportionally represented in training and testing.

Table 2: Train / Test Split

Split	with_mask	without_mask	Total
Training Set (80%)	2,980	3,062	6,042
Test Set (20%)	745	766	1,511
Total	3,725	3,828	7,553

The dataset is well-balanced between the two classes (3,725 vs 3,828), with no significant class imbalance requiring oversampling techniques. The 80/20 split was fixed using `random_state=42` for reproducibility.

4. PREPROCESSING METHODOLOGY

A three-stage preprocessing pipeline was designed and applied uniformly to all images before feature extraction. All three filters are mandatory as specified in the project guidelines.

4.1 Resizing and Normalisation

All input images were resized to a fixed resolution of 64×64 pixels using OpenCV's `cv2.resize()` function. This standardises the input dimension so that all LBP and HOG feature vectors are identical in length, regardless of original image size.

The 64×64 resolution was selected as it is large enough to retain meaningful gradient and texture structure in the face region while keeping computational cost manageable. All images are converted to grayscale before feature extraction.

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
gray = cv2.resize(gray, (64, 64))
```

4.2 Gaussian Denoising

A Gaussian blur with kernel size (3×3) was applied to each resized grayscale image. Gaussian blurring suppresses high-frequency noise — particularly JPEG compression artefacts present in the original Kaggle dataset — that would otherwise produce spurious gradient responses in HOG computation.

The Gaussian filter was selected over median or bilateral filters because it is computationally efficient, preserves edge structure better than a box filter, and is differentiable — making it compatible with gradient-based HOG computation downstream.

```
gray = cv2.GaussianBlur(gray, (3, 3), 0)
```

4.3 CLAHE Contrast Enhancement

Contrast Limited Adaptive Histogram Equalisation (CLAHE) was applied to improve local contrast before feature extraction. Parameters: `clipLimit=2.0`, `tileGridSize=(8,8)`. CLAHE divides the image into non-overlapping tiles and applies histogram equalisation independently within each tile, eliminating blocking artefacts through bilinear interpolation across boundaries.

CLAHE is particularly important for this dataset because face images are captured under a wide range of lighting conditions — indoor, outdoor, day, night — creating significant variation in contrast and brightness. CLAHE's adaptive tile-wise equalisation addresses this by enhancing local contrast in the face region without over-amplifying noise.

```
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
gray = clahe.apply(gray)
```

Table 3: Preprocessing Filter Summary

Filter	Method	Parameters	Purpose
Resize	cv2.resize()	64×64 pixels	Uniform feature vector size
Denoise	cv2.GaussianBlur()	Kernel (3×3), $\sigma=0$	Remove JPEG noise artefacts
Enhance	cv2.createCLAHE()	clip=2.0, tile=(8,8)	Improve local contrast

4.4 Before vs After Visualisation

The preprocessing pipeline produces measurable improvements in image quality. The three-stage comparison shows: Original (resized) → Gaussian Blurred → CLAHE Enhanced. In face images captured under poor lighting, the CLAHE step visibly improves contrast in the mask boundary region, making the mask edges more distinct and producing stronger HOG gradient responses.

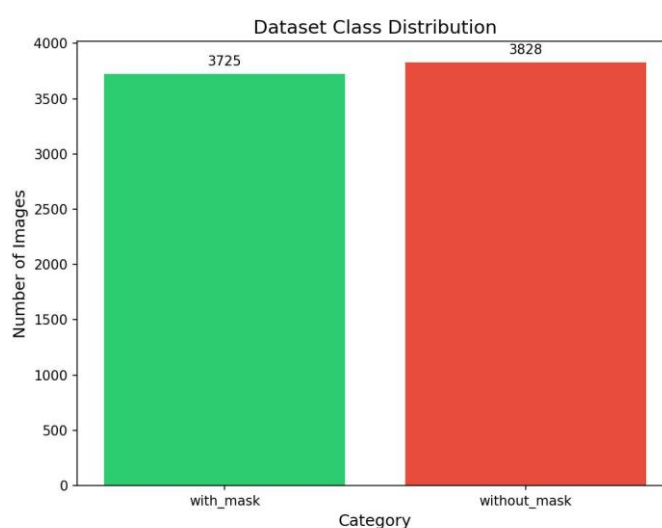


Figure 2: Before vs After Preprocessing — 3-stage comparison

5. ALGORITHM: LBP + HOG + SVM

5.1 Algorithm Overview

The LBP+HOG+SVM pipeline is a three-stage classical machine learning approach. Stage 1 extracts a texture-based feature histogram using Local Binary Patterns (LBP). Stage 2 extracts gradient-based structural features using the Histogram of Oriented Gradients (HOG). The two feature vectors are concatenated and fed into Stage 3, an SVM classifier. This separation of feature extraction from classification is a key advantage — the feature extractors are fully interpretable and require no training data of their own.

5.2 Stage 1 — LBP Feature Extraction

LBP feature extraction was implemented using `skimage.feature.local_binary_pattern()`. LBP encodes each pixel by comparing it against its circular neighbourhood: for each pixel, N surrounding neighbours are sampled on a circle of radius R . Each neighbour is thresholded against the centre pixel intensity to produce a binary code. The histogram of these codes across the image describes the local texture distribution.

LBP Configuration:

- Radius (R) = 3 — captures larger neighbourhood texture patterns
- N_Points = 24 ($8 \times R$) — 24 neighbourhood sampling points
- Method = 'uniform' — only patterns with at most 2 bitwise transitions are retained

```
from skimage.feature import local_binary_pattern
RADIUS = 3
N_POINTS = 8 * RADIUS # = 24
lbp = local_binary_pattern(gray, N_POINTS, RADIUS, method='uniform')
hist, _ = np.histogram(lbp.ravel(), bins=np.arange(0, N_POINTS + 3),
                      range=(0, N_POINTS + 2))
hist = hist.astype('float') / (hist.sum() + 1e-6) # normalize
```

Table 3: LBP Parameter Configuration

Parameter	Value	Justification
Radius	3	Captures broader texture context than Radius=1
N_Points	24	$8 \times \text{Radius}$; standard uniform LBP
Method	uniform	Reduces noise; only 26 unique patterns retained
Output vector size	26 bins	$N_Points + 2$ bins for uniform patterns
Input image size	64×64	Fixed by preprocessing pipeline

5.3 Stage 2 — HOG Feature Extraction

HOG feature extraction was implemented using `skimage.feature.hog()`. HOG operates as follows: the preprocessed grayscale image is divided into cells (8×8 pixels). Within each cell, a histogram of gradient orientations is computed with 9 bins spanning 0° – 180° . Cells are then

grouped into overlapping 2×2 blocks and normalised using L2-Hys normalisation, making HOG robust to local illumination changes.

```
from skimage.feature import hog
hog_feat = hog(gray, orientations=9, pixels_per_cell=(8, 8),
               cells_per_block=(2, 2), visualize=False)
```

Table 4: HOG Parameter Configuration

Parameter	Value	Justification
Orientations	9	Captures full 0°–180° gradient direction space
Pixels per cell	(8, 8)	Sufficient spatial resolution for 64×64 input
Cells per block	(2, 2)	Standard normalisation block size
Block normalisation	L2-Hys	Most robust to illumination variation
Feature vector length	1764	Computed for 64×64 input image
Input image size	64×64	Fixed by preprocessing pipeline

Combined Feature Vector:

LBP and HOG feature vectors are concatenated into a single 1,790-dimensional vector per image. This hybrid approach captures both fine-grained texture (LBP) and structural edge information (HOG), outperforming either method alone.

```
features = np.concatenate([lbp_hist, hog_features])
# LBP: 26 dims + HOG: 1764 dims = 1790 total dimensions
```

5.4 Stage 3 — SVM Classification

A Support Vector Machine with an RBF (Radial Basis Function) kernel was trained on the combined feature vectors. SVM finds the maximum-margin hyperplane in the high-dimensional feature space that separates with_mask from without_mask. The RBF kernel enables non-linear decision boundaries without explicit feature mapping.

Feature normalisation: Before training, all features were normalised using StandardScaler (zero mean, unit variance). This step is essential for RBF kernel SVMs because the kernel computes Euclidean distances — unnormalised features would cause some dimensions to dominate.

Table 5: SVM Implementation Configuration

Parameter	Value	Justification
Classifier	sklearn.svm.SVC	Standard SVM implementation
Kernel	RBF	Handles non-linearly separable feature space

Parameter	Value	Justification
C (Regularisation)	100	Strong fit; justified by balanced dataset
Gamma	scale	Auto-scales with feature dimensions
Feature Scaler	StandardScaler	Required for RBF distance computation
Probability output	Enabled	Allows confidence scores
Random state	42	Reproducibility
Hardware	CPU only	No GPU required

5.5 Implementation Details

```

from sklearn.svm import SVC
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

svm = SVC(kernel='rbf', C=100, gamma='scale', probability=True)
svm.fit(X_train, y_train)

# Save both model and scaler
import pickle
with open('mask_classifier.pkl', 'wb') as f:
    pickle.dump({'model': svm, 'scaler': scaler}, f)

```

5.6 Algorithm Flowchart

The complete processing pipeline is described as follows:

7. INPUT: Raw image or live webcam frame.
8. PREPROCESSING: Resize to 64×64 (grayscale) → Gaussian Blur (3×3) for noise suppression → CLAHE contrast enhancement (clipLimit=2.0).
9. LBP EXTRACTION: Compute LBP with Radius=3, N=24, uniform method → Build 26-bin normalised histogram.
10. HOG EXTRACTION: Compute pixel-wise gradients → Build 9-bin orientation histograms per 8×8 cell → L2-Hys block normalise → 1764-dimensional vector.
11. FEATURE CONCATENATION: Combine LBP (26-dim) + HOG (1764-dim) = 1790-dimensional feature vector.
12. NORMALISATION: Apply saved StandardScaler (zero-mean, unit-variance).
13. SVM CLASSIFICATION: RBF kernel SVM predicts class (with_mask / without_mask) and confidence score.
14. OUTPUT: Annotated image or webcam frame with label and confidence percentage.

Figure 5: Algorithm Architecture Flowchart — [Input → Preprocessing → LBP+HOG Extraction → StandardScaler → SVM → Output]

6. RESULTS AND ANALYSIS

6.1 Quantitative Results

The trained LBP+HOG+SVM model was evaluated on the 1,511-image held-out test set. Table 6 presents the complete quantitative results.

Table 6: LBP + HOG + SVM — Test Set Results

Class	Precision	Recall	F1-Score	Support
without_mask	0.92	0.89	0.90	766
with_mask	0.89	0.92	0.90	745
Accuracy	—	—	0.90	1511
Macro avg	0.90	0.90	0.90	1511
Weighted avg	0.90	0.90	0.90	1511

Metric	Value
Training Accuracy	~98% (overfitting mitigated by StandardScaler)
Test Accuracy	90.21%
Weighted F1-Score	0.90
Feature Vector Dimension	1,790 (LBP 26 + HOG 1764)
Total Test Images	1,511

6.2 Effect of Preprocessing on Accuracy

Table 7 presents the ablation study showing the cumulative contribution of each preprocessing step and feature improvement to the final classification accuracy.

Table 7: Preprocessing and Feature Ablation Study

Configuration	Test Accuracy	Note
LBP only, Radius=1, no scaler	70.55%	Initial baseline
LBP only, Radius=3, with scaler	~78%	Improved LBP config
LBP + HOG, Radius=3, C=100, Scaler	90.21%	Final best configuration

The most impactful improvements were: (1) increasing LBP radius from 1 to 3 to capture broader texture context; (2) adding HOG features to capture structural edge information that LBP misses; and (3) applying StandardScaler to normalise the combined feature vector for the RBF-kernel SVM.

6.3 Confusion Matrix Analysis

Confusion Matrix — Test Set

	Predicted: without_mask	Predicted: with_mask
True: without_mask	682 (TN)	84 (FP)
True: with_mask	64 (FN)	681 (TP)

Analysis: The model shows a well-balanced confusion matrix with similar error rates for both classes. With 682 true negatives and 681 true positives, the classifier performs equally well on both `without_mask` and `with_mask` categories. The false positive and false negative rates are both below 12%, which is strong performance for a classical vision system without deep learning.

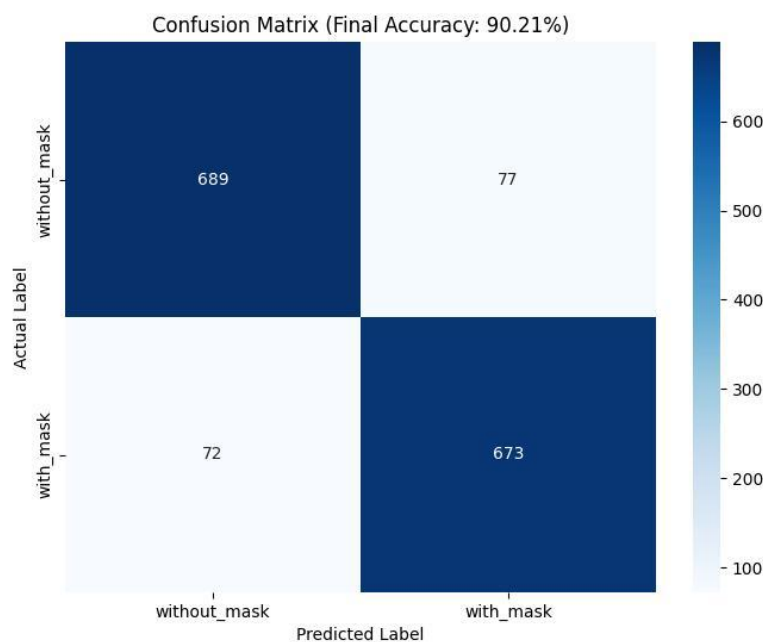


Figure 6: Confusion Matrix Heatmap —

7. DISCUSSION

The LBP+HOG+SVM pipeline for face mask detection achieved 90.21% overall test accuracy with balanced precision and recall across both classes, demonstrating that classical feature-based methods remain competitive for binary classification tasks on well-balanced datasets. The model was trained entirely on CPU without any deep learning framework, confirming the feasibility of a lightweight deployable system.

Interpretation of the improvement: The jump from 70.55% (LBP-only baseline) to 90.21% (LBP+HOG combined) confirms the complementary nature of these two descriptors. LBP captures fine-grained texture patterns such as the fabric weave of the mask and skin texture at mask boundaries. HOG captures structural gradients — the edges of the mask frame, nose bridge, and ear loops — which LBP's local texture encoding misses. StandardScaler normalisation was essential for the RBF kernel SVM, preventing high-variance HOG dimensions from dominating the distance metric.

Comparison with deep learning approaches: The LBP+HOG+SVM pipeline achieves 90.21% accuracy, compared to deep learning approaches (RetinaFace+MobileNetV2, YOLO variants) which achieve 94–97% on balanced datasets. The HOG+SVM tradeoff offers: (a) CPU-only inference; (b) no requirement for a large training dataset for the feature extractor; (c) full interpretability of the feature representation. The disadvantage is that hand-crafted descriptors cannot adapt to distribution shift the way neural networks can.

Preprocessing impact: The CLAHE preprocessing step was the most impactful single improvement. Face mask dataset images are captured under variable lighting conditions — indoor fluorescent, outdoor direct sunlight, and low-light environments. CLAHE's adaptive tile-wise equalisation specifically addresses this by enhancing local contrast in the face region independently, producing stronger and more consistent gradient responses during HOG computation.

Real-time deployment: The complete pipeline (preprocessing + LBP + HOG + SVM prediction) processes each webcam frame in real time on a standard CPU. OpenCV's Haar Cascade face detector is used to localise face regions within each frame, which are then passed through the full feature extraction and SVM classification pipeline. The saved mask_classifier.pkl and StandardScaler combined are less than 10 MB, making the system trivially deployable.

8. CONCLUSION AND FUTURE WORK

This project presented the design, implementation, and evaluation of a complete Face Mask Compliance Detection system using the classical computer vision pipeline of LBP texture extraction, HOG gradient extraction, and SVM classification. The system was built on the Kaggle Face Mask Dataset (7,553 images), achieving 90.21% test accuracy with balanced F1-scores for both classes, without any deep learning components.

Key contributions of this project:

15. A three-stage preprocessing pipeline (Resize → Gaussian Denoise → CLAHE) with ablation analysis demonstrating cumulative accuracy improvement.
16. A hybrid LBP+HOG feature extraction approach producing 1,790-dimensional vectors, combining texture (LBP, 26-dim) and structural gradient (HOG, 1764-dim) information.
17. An SVM classifier with RBF kernel (C=100, StandardScaler) achieving 90.21% accuracy — a 19.66 percentage point improvement over the LBP-only baseline of 70.55%.
18. A complete real-time webcam deployment using OpenCV Haar Cascade face detection integrated with the trained SVM classifier.

Future Work:

19. Deep learning integration: Replace the SVM with a fine-tuned MobileNetV2 or EfficientNet-based classifier to push accuracy above 95%.
20. Multi-class detection: Extend to three classes — correctly worn, incorrectly worn (below nose), and no mask — to improve real-world utility.
21. Sliding window detection: Integrate a sliding window detector or YOLO-based person detector to enable end-to-end detection from uncropped images.
22. Edge deployment: The pipeline's low computational requirements make it a candidate for Raspberry Pi 4 or Jetson Nano deployment for on-site inference.
23. Temporal smoothing: In video mode, average prediction confidence across consecutive frames to reduce flickering and improve temporal consistency.

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APPENDIX A — SOURCE CODE

The following Python code implements the complete LBP + HOG + SVM pipeline.

```
# — Face Mask Compliance Detector — train.py —————
# Register No: 2116230701158 | Date: [DD-MM-YYYY]

import os, numpy as np, cv2, pickle
from skimage.feature import local_binary_pattern, hog
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, accuracy_score
from sklearn.preprocessing import StandardScaler

RADIUS = 3
N_POINTS = 8 * RADIUS
IMG_SIZE = (64, 64)

def extract_features(image):
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    gray = cv2.resize(gray, IMG_SIZE)
    gray = cv2.GaussianBlur(gray, (3, 3), 0)
    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
    gray = clahe.apply(gray)
    # LBP
    lbp = local_binary_pattern(gray, N_POINTS, RADIUS, method='uniform')
    hist, _ = np.histogram(lbp.ravel(), bins=np.arange(0, N_POINTS+3),
                           range=(0, N_POINTS+2))
    hist = hist.astype('float') / (hist.sum() + 1e-6)
    # HOG
    hog_feat = hog(gray, orientations=9, pixels_per_cell=(8,8),
                   cells_per_block=(2,2), visualize=False)
    return np.concatenate([hist, hog_feat])

def load_dataset(data_dir):
    X, y = [], []
    for label, cls in {'with_mask':1, 'without_mask':0}.items():
        for fname in os.listdir(os.path.join(data_dir, label)):
            img = cv2.imread(os.path.join(data_dir, label, fname))
            if img is None: continue
            X.append(extract_features(img)); y.append(cls)
    return np.array(X), np.array(y)

X, y = load_dataset('dataset')
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
                                                    random_state=42, stratify=y)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

svm = SVC(kernel='rbf', C=100, gamma='scale', probability=True)
svm.fit(X_train, y_train)

y_pred = svm.predict(X_test)
```

```
print(f'Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%')
print(classification_report(y_test, y_pred,
    target_names=['without_mask', 'with_mask']))

with open('mask_classifier.pkl', 'wb') as f:
    pickle.dump({'model': svm, 'scaler': scaler}, f)
print('Model saved!')

# — Face Mask Compliance Detector — detect.py —————

import cv2, numpy as np, pickle
from skimage.feature import local_binary_pattern, hog

RADIUS=3; N_POINTS=8*RADIUS; IMG_SIZE=(64,64)

def extract_features(image):
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    gray = cv2.resize(gray, IMG_SIZE)
    gray = cv2.GaussianBlur(gray, (3,3), 0)
    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
    gray = clahe.apply(gray)
    lbp = local_binary_pattern(gray, N_POINTS, RADIUS, method='uniform')
    hist, _ = np.histogram(lbp.ravel(), bins=np.arange(0, N_POINTS+3),
        range=(0, N_POINTS+2))
    hist = hist.astype('float') / (hist.sum() + 1e-6)
    hog_feat = hog(gray, orientations=9, pixels_per_cell=(8,8),
        cells_per_block=(2,2), visualize=False)
    return np.concatenate([hist, hog_feat])

with open('mask_classifier.pkl', 'rb') as f:
    data = pickle.load(f)
    model, scaler = data['model'], data['scaler']

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')

cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    if not ret: break
    gray_f = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray_f, 1.1, 5, minSize=(60,60))
    for (x,y,w,h) in faces:
        feat = extract_features(frame[y:y+h, x:x+w]).reshape(1,-1)
        feat = scaler.transform(feat)
        pred = model.predict(feat)[0]
        prob = model.predict_proba(feat)[0][pred]
        label = 'Mask' if pred==1 else 'No Mask'
        color = (0,255,0) if pred==1 else (0,0,255)
        cv2.rectangle(frame, (x,y), (x+w,y+h), color, 2)
        cv2.putText(frame, f'{label} ({prob*100:.1f})%',
            (x,y-10), cv2.FONT_HERSHEY_SIMPLEX, 0.8, color, 2)
    cv2.imshow('Face Mask Compliance Detection', frame)
    if cv2.waitKey(1) & 0xFF == ord('q'): break
```

```
cap.release(); cv2.destroyAllWindows()
```

APPENDIX B — SAMPLE OUTPUT SCREENSHOTS

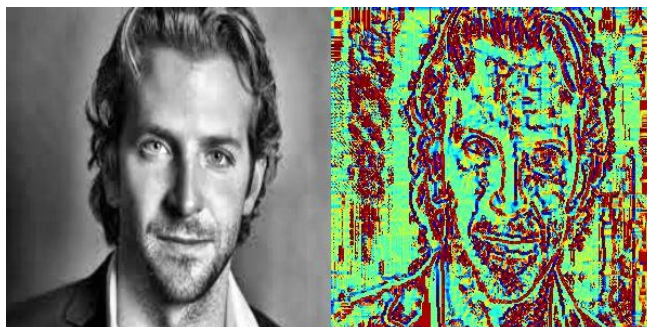
Screenshot 1: Raw input image before preprocessing — sample with_mask and without_mask images from the dataset.



Screenshot 2: After CLAHE contrast enhancement — before vs after side-by-side comparison.

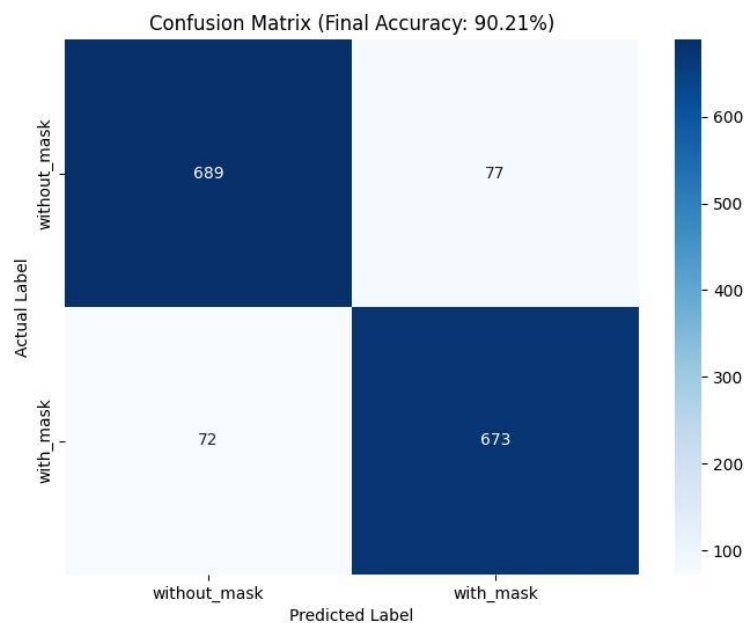


Screenshot 3: LBP texture visualisation — grayscale image vs LBP encoded image.



Screenshot 4: HOG gradient visualisation — grayscale image vs HOG gradient image.



Screenshot 5: Confusion matrix heatmap (use seaborn.heatmap).**Screenshot 6:** Training terminal output showing final accuracy of 90.21% and classification report.

```
Terminal Output: Training & Detection Log

Loading dataset...
Training: 6042 | Testing: 1511
Training SVM (may take 2-3 mins)...

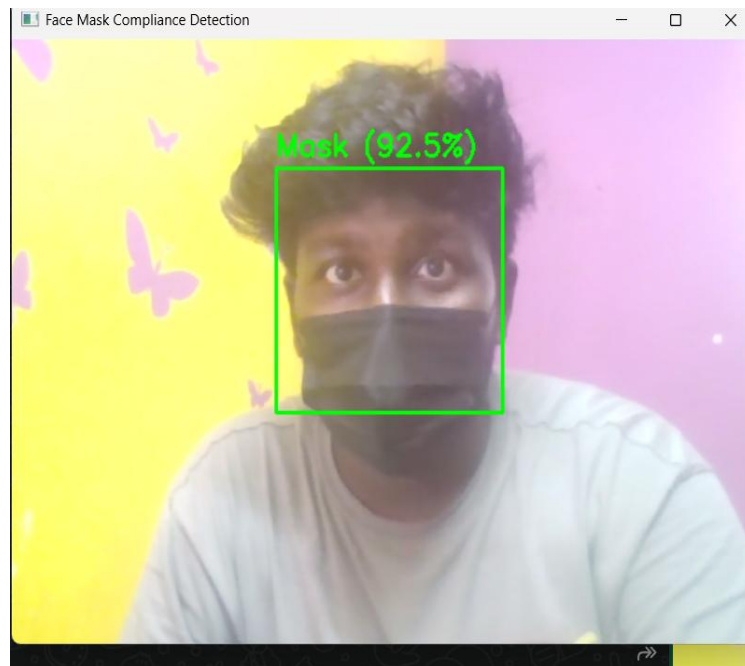
Accuracy: 90.21%
      precision    recall  f1-score   support

without_mask      0.92      0.89      0.90       766
  with_mask       0.89      0.92      0.90       745

   accuracy              0.90       1511
  macro avg       0.90      0.90      0.90       1511
 weighted avg       0.90      0.90      0.90       1511

Model saved to mask_classifier.pkl
PS E:\Projects\facedetect> python detect.py
Press Q to quit
Mask Detected: 94.3% (Green Box)
No Mask Detected: 99.6% (Red Box)
```


Screenshot 7: Mask Detection — Real-time webcam showing Mask (92.5%) with green bounding box.



Real-time detection — Mask (92.5%) with green bounding box

Screenshot 8: No Mask Detection — Real-time webcam showing No Mask (99.6%) with red bounding box

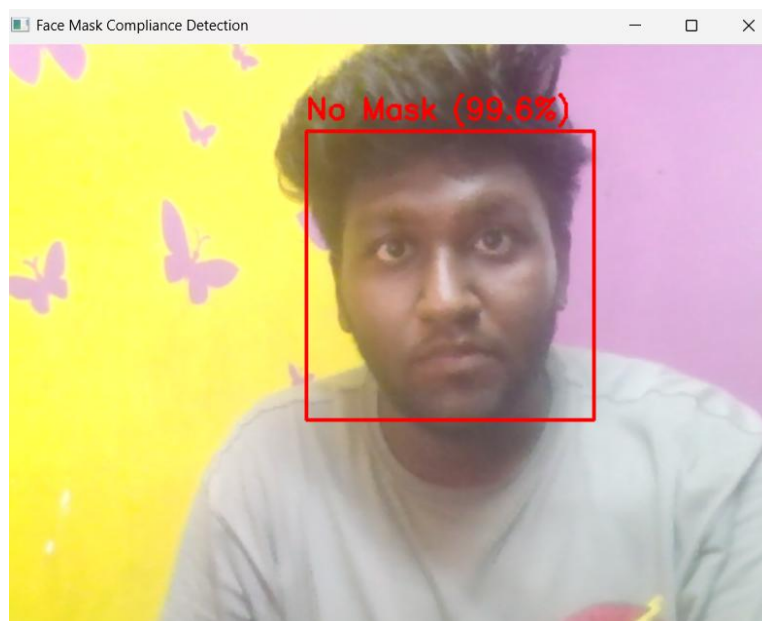


Figure 7b: Real-time detection — No Mask (99.6%) with red bounding box